

Self-Check Questions: Multivariable Calculus

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1 Multivariable Calculus

1.1 Basic Types of Functions

1. What is a scalar-valued function of a scalar variable $f : \mathbb{R} \rightarrow \mathbb{R}$?
2. What is a scalar-valued function of several variables? What is a scalar-valued function of a vector variable $f : \mathbb{R}_n \rightarrow \mathbb{R}$?
3. What is a vector-valued function of one variable? What is a vector-valued function of a scalar variable $f : \mathbb{R} \rightarrow \mathbb{R}_n$?
4. What is a vector-valued function of several variables? What is a vector-valued function of a vector variable $f : \mathbb{R}_n \rightarrow \mathbb{R}_m$?
5. What is a scalar field? What is a vector field?
6. When can two functions be composed? State the condition in terms of domains and codomains.
7. Give examples of composing: (a) a scalar-valued function with a vector-valued function, (b) two vector-valued functions.
8. What is the difference between a coordinate transformation and a vector field?

1.2 Vector-Valued Functions of One Variable (Curves)

9. How do we define the limit of a vector-valued function $\mathbf{r}(t)$?
10. How do we define continuity of a vector-valued function?
11. How do we define the derivative of $\mathbf{r}(t)$? Give the definition from first principles.
12. What is the geometric meaning of $\mathbf{r}'(t)$?
13. What is the equation of the tangent line to a curve at t_0 ?
14. State the differentiation rules (sum, scalar multiplication, dot product, cross product).
 - (a) What is the derivative of the sum of two vector-valued functions $\mathbf{r}(t)$ and $\mathbf{s}(t)$?

$$\frac{d}{dt}(\mathbf{r}(t) + \mathbf{s}(t)) ?$$

- (b) What is the derivative of a scalar function $f(t)$ times a vector-valued function $\mathbf{r}(t)$?

$$\frac{d}{dt}(f(t)\mathbf{r}(t)) ?$$

- (c) What is the derivative of the dot product of two vector-valued functions $\mathbf{r}(t)$ and $\mathbf{s}(t)$?

$$\frac{d}{dt}(\mathbf{r}(t) \cdot \mathbf{s}(t)) ?$$

- (d) What is the derivative of the cross product of two vector-valued functions $\mathbf{r}(t)$ and $\mathbf{s}(t)$?

$$\frac{d}{dt}(\mathbf{r}(t) \times \mathbf{s}(t)) ?$$

15. Prove that if $|\mathbf{r}(t)| = c$, then $\mathbf{r}(t) \cdot \mathbf{r}'(t) = 0$.

16. How do we define the integral of a vector-valued function?
17. Define the arc length of a curve.
18. What does it mean to parameterize a curve by arc length?
19. What is the unit tangent vector $\mathbf{T}$?
20. What is the unit normal vector $\mathbf{N}$? Why is it orthogonal to $\mathbf{T}$?
21. What is the unit binormal vector $\mathbf{B}$?
22. What is the osculating plane?
23. What is the normal plane?
24. What is curvature? Give its definitions.
25. What is the curvature of a line? Of a circle?
26. What are position, velocity, acceleration, and speed?
27. Decompose acceleration into tangential and normal components.
28. Give examples of purely tangential and purely normal acceleration.
29. What is torsion? What does it measure?
30. What are the units of curvature and torsion?

1.3 Functions of Several Variables (Scalar Fields)

31. What is a level set of $f : \mathbb{R}^n \rightarrow \mathbb{R}$?
32. What is a level curve? What is a level surface?
33. How can level surfaces be reduced to level curves?
34. How do we define the limit of $f : \mathbb{R}^n \rightarrow \mathbb{R}$?
35. What does it mean that a limit is independent of path?
36. How do we define continuity?
37. What is a partial derivative? Give the definition.
38. What is the geometric interpretation of partial derivatives?
39. What does it mean for a function to be differentiable?
40. What is the total differential?
41. What is the linear approximation of a function?
42. Describe composition with other functions and with a path $\mathbf{r}(t)$.
43. State the chain rule (general and along a path).
44. What are independent and dependent variables in implicit equations?
45. Given $F(x, y) = 0$, derive $\frac{dy}{dx}$ and $\frac{dx}{dy}$.
46. What is the relationship between $\frac{dy}{dx}$ and $\frac{dx}{dy}$?

1.4 Gradients, Geometry, and Optimization

47. What is the gradient ∇f ? Give its type (domain, codomain) and notation.
48. What is the geometric interpretation of the gradient?
49. Why is the gradient orthogonal to level curves? Prove it.
50. Why is the gradient orthogonal to level surfaces? Prove it.
51. What is the directional derivative?
52. How is the directional derivative related to the gradient?
53. In which direction is the directional derivative maximal?
54. What is the derivative of f along a path?
55. What is a critical point?
56. State Fermat's theorem.
57. What is the second derivative test?
58. What is the equation of a tangent line to a level curve?
59. What is the equation of a tangent plane to a level surface?
60. What is the equation of a normal line?
61. What is the equation of the tangent plane to $z = f(x, y)$?

1.5 Implicit Functions and Constrained Optimization

62. What problem does the Implicit Function Theorem solve?
63. State the Implicit Function Theorem for $F(x, y) = 0$.
64. What condition ensures that y can be written as a function of x ?
65. Derive $\frac{dy}{dx} = -\frac{F_x}{F_y}$.
66. What is the geometric meaning of $F_y \neq 0$?
67. What happens if $F_y = 0$?
68. Extend the theorem to $F(x, y, z) = 0$.
69. Express $\frac{\partial z}{\partial x}$ and $\frac{\partial z}{\partial y}$.
70. How does the theorem justify implicit differentiation?
71. Give examples where the theorem applies and fails.
72. What is the problem of constrained optimization?
73. State the method of Lagrange multipliers.
74. What system of equations must be solved?
75. Why must $\nabla f = \lambda \nabla g$?
76. What is the geometric interpretation of this condition?
77. How does the method extend to higher dimensions?
78. What happens if $\nabla g = \mathbf{0}$?

1.6 Vector-Valued Functions of Several Variables and Jacobians

79. What is a vector-valued function $F : \mathbb{R}^n \rightarrow \mathbb{R}^m$?
80. What is a vector field?
81. What is a parametric surface?
82. How do we compute tangent vectors to a parametric surface?
83. What is the tangent plane to a parametric surface?
84. What is the Jacobian matrix?
85. How is it defined in terms of partial derivatives?
86. What do the rows represent?
87. What do the columns represent?
88. When does the Jacobian reduce to the gradient?
89. What is the total derivative?
90. How does the Jacobian represent linear approximation?
91. What is the Jacobian of the identity map?
92. What is the shape of the Jacobian for $F : \mathbb{R}^n \rightarrow \mathbb{R}^m$?
93. When is the Jacobian invertible?
94. What is the Jacobian determinant?
95. What is its geometric meaning?
96. How is it used in change of variables?
97. What does it mean when it is zero?
98. State the Inverse Function Theorem (informally).
99. What condition ensures local invertibility?

1.7 Multiple Integrals

100. What is a double integral of a function $f(x, y)$ over a region R ?
101. How is the double integral defined using Riemann sums?
102. What is the geometric interpretation of a double integral?
103. What types of quantities can be computed using double integrals?
104. What is the difference between Type I and Type II regions?
105. How do we compute a double integral as an iterated integral?
106. State Fubini's theorem.
107. When can the order of integration be changed?
108. How is the area of a region computed using double integrals?
109. How is the average value of a function over a region defined?
110. What is a triple integral?

111. What is the geometric interpretation of a triple integral?
112. How can triple integrals be written as iterated integrals?
113. How is volume computed using triple integrals?
114. What are cylindrical coordinates?
115. What are spherical coordinates?
116. How do we convert between Cartesian and cylindrical coordinates?
117. How do we convert between Cartesian and spherical coordinates?
118. What is the Jacobian for cylindrical coordinates?
119. What is the Jacobian for spherical coordinates?
120. What is the change-of-variables formula for multiple integrals?
121. What is the geometric interpretation of the Jacobian determinant in integration?

2 Implicit Differentiation, Gradients, Level Curves, Related Rates

1. What does it mean geometrically for a function to satisfy $F(x, y) = c$?
2. If you move along a curve defined by $F(x, y) = c$, what is $\frac{d}{dx}F(x, y(x))$ and why?
3. How does the chain rule lead to the formula

$$\frac{\partial F}{\partial x} + \frac{\partial F}{\partial y} \frac{dy}{dx} = 0?$$

4. What does the vector $(1, \frac{dy}{dx})$ represent geometrically?
5. What is the gradient ∇F , and what information does it encode?
6. Rewrite implicit differentiation in dot product form. What does

$$\nabla F \cdot (1, \frac{dy}{dx}) = 0$$

mean geometrically?

7. Why must the gradient be perpendicular to level curves?
8. For the circle $F(x, y) = x^2 + y^2$, what is ∇F ? What direction does it point?
9. For the same circle, compute $\frac{dy}{dx}$ using implicit differentiation.
10. Verify explicitly that the gradient is orthogonal to the tangent direction for the circle.
11. If $x = x(t)$ and $y = y(t)$ satisfy $F(x, y) = c$, what is $\frac{dF}{dt}$ and why?
12. Derive the related rates equation

$$\frac{\partial F}{\partial x} \frac{dx}{dt} + \frac{\partial F}{\partial y} \frac{dy}{dt} = 0.$$

13. What does the vector $(\frac{dx}{dt}, \frac{dy}{dt})$ represent geometrically?
14. Rewrite related rates in dot product form. What does

$$\nabla F \cdot (x', y') = 0$$

mean?

15. What is the relationship between velocity vectors and level curves?
16. How are implicit differentiation and related rates mathematically similar?
17. What is a directional derivative, and how does it relate to $\nabla F \cdot \mathbf{v}$?
18. Why does $\nabla F \cdot \mathbf{v} = 0$ imply motion along a level curve?
19. In one sentence: how are gradients, level curves, implicit differentiation, and related rates connected?
20. Bonus: If a particle moves along a circle, why must its velocity be perpendicular to the radius?

3 Parametric vs Implicit Curves

3.1 Core understanding

1. What is the difference between a parametric representation and an implicit representation of a curve?
2. What does each representation emphasize: generation of points or membership of points?

3.2 Parametric $\rightarrow$ Implicit

3. Given $x = x(t)$ and $y = y(t)$, what does it mean to eliminate the parameter?
4. Can every parametric curve be written in the form $F(x, y) = 0$? What assumptions are needed?
5. What are examples where eliminating the parameter is:
 - easy?
 - difficult or impossible in closed form?
6. Why might the resulting implicit equation not be unique?

3.3 Implicit $\rightarrow$ Parametric

7. What does it mean to parametrize a curve defined by $F(x, y) = 0$?
8. What condition on F guarantees a local parametrization?
9. How does the Implicit Function Theorem justify this?
10. What role does the condition $\nabla F \neq 0$ play?

3.4 Local vs Global behavior

11. What is the difference between a local and a global parametrization?
12. Why can an implicit curve be parametrized locally but not globally?

3.5 Failure modes / limitations

13. Why can't some implicit curves be written as a single function $y = f(x)$?
14. What issues arise from:
 - multiple branches?
 - self-intersections?
 - cusps?
15. Can a curve with multiple disconnected components be described by a single parametrization?

3.6 Conceptual insight

16. Why is it easier to go from parametric to implicit than the other way around?
17. How does an implicit equation define a level set?
18. How do these ideas connect to level curves and gradients?

3.7 Deeper reflection

19. Can you think of a curve that:
 - has a simple parametrization but a complicated implicit form?
 - has a simple implicit form but requires multiple parametrizations?
20. What does this tell you about the trade-offs between representations?

4 Tangent Planes

1. Compare the tangent line in single-variable calculus with the tangent plane in multivariable calculus.
2. What is the definition of a tangent plane to a surface $z = f(x, y)$ at a point (x_0, y_0) ?
3. How do the partial derivatives $\frac{\partial f}{\partial x}$ and $\frac{\partial f}{\partial y}$ relate to the slope of the tangent plane?
4. Compute the tangent plane to $f(x, y) = x^2 + y^2$ at the point $(1, 2)$.
5. What is the geometric meaning of the linear approximation

$$L(x, y) = f(x_0, y_0) + f_x(x_0, y_0)(x - x_0) + f_y(x_0, y_0)(y - y_0)?$$

6. How is the tangent plane related to the total derivative of $f(x, y)$?
7. Explain how the gradient vector $\nabla f(x_0, y_0)$ is connected to the tangent plane.
8. For a surface given implicitly by $F(x, y, z) = 0$, how do you find the equation of the tangent plane at a point?
9. What role does the gradient $\nabla F(x_0, y_0, z_0)$ play in the implicit surface case?
10. Compute the tangent plane to the surface $x^2 + y^2 + z^2 = 9$ at the point $(1, 2, 2)$.
11. What is the normal vector to the tangent plane of $z = f(x, y)$?
12. How can you express the tangent plane using a normal vector and a point?
13. What is the relationship between the tangent plane and directional derivatives?
14. How can you use the tangent plane to approximate the value of a function near a point?
15. Under what circumstances might a function fail to have a tangent plane at a point?
16. Given a parameterized surface $\mathbf{r}(u, v)$, how do you compute the tangent plane at a point?
17. What vectors span the tangent plane in the parameterized surface case?
18. Compute the tangent plane to the parameterized surface $\mathbf{r}(u, v) = (u, v, u^2 + v^2)$ at $(1, 1, 2)$.
19. How is the cross product used in finding the normal vector to a parameterized surface?
20. What is the difference between a tangent plane and a secant plane?
21. How does the tangent plane behave near critical points where $\nabla f = 0$?
22. Why is the tangent plane considered the best linear approximation of a surface near a point?

5 Directional Derivatives

1. What is the definition of the directional derivative of a function $f(x, y)$ at a point (x_0, y_0) in the direction of a vector $\mathbf{v}$?
2. Why must the direction vector be normalized when computing a directional derivative?
3. Write the limit definition of the directional derivative.
4. How is the directional derivative related to the gradient $\nabla f(x_0, y_0)$?
5. State the formula for the directional derivative in terms of the gradient and a unit vector $\mathbf{u}$.
6. What is the geometric interpretation of the directional derivative?
7. Compute the directional derivative of $f(x, y) = x^2 + y^2$ at $(1, 2)$ in the direction of $\mathbf{v} = (3, 4)$.
8. What is the direction of maximum increase of a function at a point?
9. What is the maximum value of the directional derivative at a point, and how is it computed?
10. In what direction is the directional derivative equal to zero?
11. How are level curves of a function related to directional derivatives?
12. Explain why the gradient is perpendicular to level curves.
13. How can directional derivatives be used to test whether a function is constant along a curve?
14. What is the relationship between the directional derivative and the total derivative?
15. How do you compute the directional derivative for a function $f(x, y, z)$ in $\mathbb{R}^3$?
16. Compute the directional derivative of $f(x, y, z) = xyz$ at $(1, 1, 1)$ in the direction of $\mathbf{v} = (1, 1, 1)$.
17. What happens if the gradient at a point is the zero vector?
18. Can a function have all directional derivatives at a point but still fail to be differentiable there?
19. How is the directional derivative related to partial derivatives?
20. Show how the directional derivative can be expressed as a dot product.
21. What is the difference between directional derivatives and partial derivatives?
22. How does the chain rule relate to directional derivatives?
23. Given a parameterized curve $\mathbf{r}(t)$, how do you compute the rate of change of f along the curve?
24. Explain the concept of the derivative of f along a path.
25. How can directional derivatives be interpreted as slopes of tangent lines to curves on surfaces?

6 Total Derivatives

6.1 Type, Domain, and Codomain

1. Let $f : \mathbb{R}^n \rightarrow \mathbb{R}^m$. What is the type (domain and codomain) of the total derivative $Df(\mathbf{x})$ at a point $\mathbf{x} \in \mathbb{R}^n$?
2. How can you interpret $Df(\mathbf{x})$ as a linear map between tangent spaces?
3. In coordinates, how is $Df(\mathbf{x})$ represented as a matrix? What are its dimensions?

6.2 Jacobian as a Limit (First Principles)

4. Starting from the definition of differentiability, write the limit definition of $Df(\mathbf{x})$.
5. Show that if f is differentiable at $\mathbf{x}$, then there exists a matrix $J_f(\mathbf{x})$ such that

$$\lim_{\|\mathbf{h}\| \rightarrow 0} \frac{\|f(\mathbf{x} + \mathbf{h}) - f(\mathbf{x}) - J_f(\mathbf{x})\mathbf{h}\|}{\|\mathbf{h}\|} = 0.$$

6. How do the entries of $J_f(\mathbf{x})$ arise from partial derivatives?
7. Why does this limit definition guarantee that the Jacobian gives the *best linear approximation*?

6.3 Directional Derivative and Gradient

8. How does the total derivative differ from a partial derivative?
9. How does the total derivative relate to the gradient ∇f ?
10. What happens to the total derivative if one of the variables is held constant?
11. How is the total derivative related to directional derivatives?
12. Define the directional derivative of f at $\mathbf{x}$ in the direction $\mathbf{v}$ using a limit.
13. Show that the directional derivative can be expressed as

$$Df(\mathbf{x})[\mathbf{v}] = J_f(\mathbf{x})\mathbf{v}.$$

14. In the scalar-valued case $f : \mathbb{R}^n \rightarrow \mathbb{R}$, how is the Jacobian related to the gradient $\nabla f(\mathbf{x})$?
15. Why does the gradient point in the direction of steepest ascent?

6.4 Tangent Planes and Local Linearization

16. For $f : \mathbb{R}^n \rightarrow \mathbb{R}$, write the equation of the tangent hyperplane at $\mathbf{x}_0$.
17. How does the linear map $Df(\mathbf{x}_0)$ define this hyperplane?
18. What is the first-order Taylor approximation of f near $\mathbf{x}_0$?
19. Why is $Df(\mathbf{x}_0)$ called the *local linearization* of f ?

6.5 Normal Vector

20. For a level set $f(\mathbf{x}) = c$, what vector is normal to the surface at $\mathbf{x}$?
21. How is this normal vector expressed in terms of the Jacobian?
22. In the case $f : \mathbb{R}^2 \rightarrow \mathbb{R}$, how does ∇f relate to the tangent line?

6.6 Linear Map and Best Linear Approximation

23. What does it mean for $Df(\mathbf{x})$ to be a linear map?
24. What exactly is being mapped by $Df(\mathbf{x})$?
25. Why is $Df(\mathbf{x})$ the *best* linear approximation to f near $\mathbf{x}$?
26. How does the error behave compared to $\|\mathbf{h}\|$?

6.7 Jacobian Determinant and Change of Variables

27. For $f : \mathbb{R}^n \rightarrow \mathbb{R}^n$, what is the Jacobian determinant $\det J_f(\mathbf{x})$?
28. How does $\det J_f(\mathbf{x})$ describe local volume (or area) scaling?
29. Why does $|\det J_f(\mathbf{x})|$ appear in the change of variables formula for multiple integrals?
30. What happens geometrically when $\det J_f(\mathbf{x}) = 0$?

6.8 Multivariable Chain Rule

31. Let $f : \mathbb{R}^n \rightarrow \mathbb{R}^m$ and $g : \mathbb{R}^m \rightarrow \mathbb{R}^k$. State the chain rule for $D(g \circ f)(\mathbf{x})$.
32. Show that

$$J_{g \circ f}(\mathbf{x}) = J_g(f(\mathbf{x})) J_f(\mathbf{x}).$$

33. How does this relate to composition of linear maps?

6.9 Connection to Taylor Series

34. Write the first-order Taylor expansion of $f : \mathbb{R}^n \rightarrow \mathbb{R}^m$ at $\mathbf{x}_0$.
35. Where does the Jacobian appear in this expansion?
36. How would higher-order derivatives extend this approximation?

6.10 Newton's Method for Systems

37. Let $F : \mathbb{R}^n \rightarrow \mathbb{R}^n$. Write the Newton iteration for solving $F(\mathbf{x}) = \mathbf{0}$.
38. How is the Jacobian matrix used in each iteration?
39. Why does Newton's method rely on local linearization?
40. What conditions on $J_F(\mathbf{x})$ are required for the method to work?

6.11 Mapping Tangent Vectors

41. Let $f : \mathbb{R}^n \rightarrow \mathbb{R}^m$ and let $\gamma : \mathbb{R} \rightarrow \mathbb{R}^n$ be a differentiable curve with $\gamma(0) = x_0$. What is the type (domain and codomain) of $\gamma'(0)$ and of $Df(x_0)\gamma'(0)$?
42. Given a curve $\gamma(t)$ with $\gamma(0) = x_0$, write the expression for the tangent vector to the image curve $f(\gamma(t))$ at $t = 0$. Which theorem justifies this expression?
43. State and explain the identity

$$\left. \frac{d}{dt} f(\gamma(t)) \right|_{t=0} = Df(x_0) \gamma'(0).$$

What are the roles of each term in this equation?

44. Why does the Jacobian $Df(x_0)$ depend only on the tangent vector $\gamma'(0)$ and not on the specific curve γ ? What does this imply about its geometric meaning?
45. Suppose two curves $\gamma_1(t)$ and $\gamma_2(t)$ satisfy $\gamma_1(0) = \gamma_2(0) = x_0$ and $\gamma_1'(0) = \gamma_2'(0)$. What can you say about the derivatives of their image curves under f at $t = 0$?
46. Show that for any vector $v \in \mathbb{R}^n$, the curve $\gamma(t) = x_0 + tv$ satisfies

$$Df(x_0)v = \left. \frac{d}{dt} f(x_0 + tv) \right|_{t=0}.$$

What does this tell you about the relationship between the Jacobian and directional derivatives?

47. In what sense can the Jacobian $Df(x_0)$ be viewed as a map between tangent spaces? Write explicitly the mapping it defines.
48. Explain what is meant by the statement: “the Jacobian maps tangent vectors at x_0 to tangent vectors at $f(x_0)$.” What are the spaces involved?
49. Consider all curves passing through x_0 . What structure do their tangent vectors at $t = 0$ form? How does the Jacobian act on this structure?
50. Why is the Jacobian called the “best linear approximation” of f at x_0 ? How does the curve-based interpretation support this claim?
51. Let $f : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ and let $\gamma(t)$ be a curve through x_0 . Describe geometrically how the tangent vector $\gamma'(0)$ is transformed by $Df(x_0)$.
52. How does the mapping of tangent vectors by $Df(x_0)$ lead to the notion of a tangent line, plane, or hyperplane to the image of f ?
53. If $f : \mathbb{R}^n \rightarrow \mathbb{R}$, how does $Df(x_0)$ relate to the gradient $\nabla f(x_0)$? How does this affect the interpretation of $Df(x_0)v$?
54. Explain why the Jacobian provides a linear map that approximates how small displacements near x_0 are transformed under f .

7 The D Notation

1. What is the domain and codomain of the function $f(x, y) = x^2 + y^2$?
2. What is the domain and codomain of the derivative $Df(x, y)$?
3. In general, for a function $f : \mathbb{R}^m \rightarrow \mathbb{R}^n$, what is the type (domain and codomain) of $Df(x)$?
4. Why is $Df(x, y)$ considered a linear map rather than just a collection of partial derivatives?
5. Compute the partial derivatives $\frac{\partial f}{\partial x}$ and $\frac{\partial f}{\partial y}$ for $f(x, y) = x^2 + y^2$.
6. Write the gradient $\nabla f(x, y)$.
7. Express $Df(x, y)$ as a row vector (Jacobian matrix).
8. Let $h = (h_1, h_2) \in \mathbb{R}^2$. Compute $Df(x, y)(h)$.
9. How does the expression $Df(x, y)(h)$ relate to the dot product with the gradient?
10. Write the first-order linear approximation of $f(x, y)$ at a point (x, y) using $Df(x, y)$.
11. What is the geometric meaning of the vector $\nabla f(x, y)$?
12. What does $Df(x, y)(h)$ represent geometrically?
13. How does the definition of $Df(x, y)(h)$ relate to the directional derivative?
14. Compare the expressions:

$$Df(x)(h) = 2x \cdot h \quad \text{and} \quad Df(x, y)(h) = 2x \cdot h_1 + 2y \cdot h_2.$$

What structural similarity do you observe?

15. In the scalar-to-scalar case $f : \mathbb{R} \rightarrow \mathbb{R}$, why can $Df(x)$ be identified with multiplication by a number?
16. Explain why $Df(x, y)$ can be viewed as the “best linear approximation” of f near (x, y) .
17. If $h = (1, 0)$, what does $Df(x, y)(h)$ compute? What about $h = (0, 1)$?
18. How would you interpret $Df(x, y)$ acting on an arbitrary direction vector h in terms of rates of change?

Scalar $\rightarrow$ Scalar	$f : \mathbb{R} \rightarrow \mathbb{R}$	$Df(x)(h) = f'(x) h$
Scalar $\rightarrow$ Vector	$f : \mathbb{R} \rightarrow \mathbb{R}^n$	$Df(x)(h) = f'(x) \cdot h$
Vector $\rightarrow$ Scalar	$f : \mathbb{R}^m \rightarrow \mathbb{R}$	$Df(x)(h) = \nabla f(x) \cdot h$
Vector $\rightarrow$ Vector	$f : \mathbb{R}^m \rightarrow \mathbb{R}^n$	$Df(x)(h) = J_f(x) h$

A hint: Think of D as a type-lifting operator that transforms a function into a new function that returns linear maps

$$D : (\mathbb{R}^m \rightarrow \mathbb{R}^n) \rightarrow (\mathbb{R}^m \rightarrow \mathcal{L}(\mathbb{R}^m, \mathbb{R}^n))$$

where $\mathcal{L}(\cdot) =$ space of linear maps.

A hint: Think of $Df(x)$ as: “Give me a direction h , and I’ll tell you how f changes in that direction.”

$$Df(x)(h) \approx f(x + h) - f(x)$$

8 Implicit Function Theorem

1. What does it mean for a function to be defined implicitly?
2. How does the implicit function theorem explain why many curves locally look like graphs of functions?
3. Give an example of an equation that defines a relationship between variables implicitly.
4. Why can some implicit equations not be solved algebraically for one variable?
5. What question does the implicit function theorem answer about such equations?
6. State the implicit function theorem for a function $F(x, y)$.
7. What conditions must the function $F(x, y)$ satisfy for the theorem to apply?
8. Why is the condition $F_y(x_0, y_0) \neq 0$ important?
9. What does the theorem guarantee about the existence of a function $y = f(x)$?
10. In what sense does the resulting function exist (global or local)?
11. What does “locally” mean in the statement of the implicit function theorem?
12. How does the theorem justify the use of implicit differentiation?
13. Derive the formula for $\frac{dy}{dx}$ from the equation $F(x, y) = 0$.
14. What geometric feature often occurs when $F_y = 0$?
15. At which points of the circle $x^2 + y^2 = 1$ does the implicit function theorem fail to produce $y = f(x)$?
16. How can the theorem be used to justify solving equations numerically?
17. How does the theorem generalize to functions $F : \mathbb{R}^{n+m} \rightarrow \mathbb{R}^m$?
18. What role does the Jacobian matrix play in the multivariable version of the theorem?
19. What rank condition must the Jacobian satisfy?
20. What does the theorem guarantee about expressing some variables as functions of others?
21. How many variables can be solved for in terms of the remaining variables?
22. How does the implicit function theorem relate to the inverse function theorem?
23. In what sense can the inverse function theorem be viewed as a special case of the implicit function theorem?
24. How is the implicit function theorem used in constrained optimization?
25. Why is the theorem important for the method of Lagrange multipliers?
26. How does the theorem help describe surfaces defined by equations $F(x, y, z) = 0$?
27. What geometric object is typically formed by the solution set of such an equation?
28. What does the theorem imply about the smoothness of these solution sets?
29. What would go wrong in calculus if the implicit function theorem were not true?

9 Change of Variables in Multiple Integrals

9.1 Conceptual Foundations

1. What is the general idea behind a change of variables in a double integral?
2. What is meant by a transformation

$$T(u, v) = (x(u, v), y(u, v))?$$

3. What conditions must a transformation $T(u, v)$ satisfy in order for the change of variables theorem to apply?
4. What is the Jacobian matrix of a transformation $T(u, v) = (x(u, v), y(u, v))$?
5. Define the Jacobian determinant:

$$J(u, v) = \frac{\partial(x, y)}{\partial(u, v)}.$$

6. Write the explicit formula for the Jacobian determinant in terms of partial derivatives.
7. Why does the absolute value of the Jacobian determinant appear in the change-of-variables formula?
8. What geometric meaning does the Jacobian determinant have?
9. What does it mean if the Jacobian determinant is zero at a point?
10. What happens to area elements under a transformation?

9.2 Change of Variables Theorem (Double Integrals)

11. State the Change of Variables Theorem for double integrals.
12. Complete the formula:

$$\iint_R f(x, y) dA = \iint_S f(x(u, v), y(u, v)) \left| \frac{\partial(x, y)}{\partial(u, v)} \right| du dv.$$

13. What is the relationship between the regions R and S ?
14. Why is it often easier to integrate over S rather than R ?
15. Describe the general steps required to compute a double integral using change of variables.

9.3 Computing Jacobians

16. Compute the Jacobian determinant of the transformation:

$$x = u^2 - v^2, \quad y = 2uv.$$

17. Compute the Jacobian determinant for:

$$x = au + bv, \quad y = cu + dv.$$

18. For what condition on a, b, c, d is the transformation invertible?
19. How does the Jacobian determinant behave under composition of transformations?
20. If T has Jacobian J_T and its inverse has Jacobian $J_{T^{-1}}$, what is the relationship between them?

9.4 Polar Coordinates

21. Write the transformation from polar to Cartesian coordinates.
22. Compute the Jacobian determinant for the polar transformation:

$$x = r \cos \theta, \quad y = r \sin \theta.$$

23. Why does the area element become

$$dA = r \, dr \, d\theta$$

in polar coordinates?

24. Rewrite the integral

$$\iint_R f(x, y) \, dA$$

in polar coordinates.

25. Describe a region in the plane that is particularly well suited for polar coordinates.
26. Evaluate conceptually:

$$\iint_R (x^2 + y^2) \, dA$$

over a disk using polar coordinates.

9.5 Geometric Interpretation

27. How does a small rectangle in the uv -plane transform under T ?
28. How does the Jacobian determinant approximate local area scaling?
29. What is the geometric interpretation of a negative Jacobian determinant?
30. Why do we take the absolute value in the change of variables formula?

9.6 Strategy and Problem Solving

31. When facing a difficult region R , how do you decide on a good transformation?
32. Why are linear transformations often useful for parallelogram-shaped regions?
33. How can you transform an elliptical region into a circular region?
34. What is the advantage of transforming complicated boundaries into rectangular ones?
35. What common algebra mistakes occur when computing Jacobians?
36. Why is it important to rewrite the integrand entirely in terms of u and v ?
37. What must always be changed: the integrand, the differential, the region?
38. How can you check whether your transformation is one-to-one?

9.7 Extension to Triple Integrals

- 39. State the change of variables formula for triple integrals.
- 40. Write the definition of the Jacobian determinant for a transformation

$$(u, v, w) \mapsto (x, y, z).$$

- 41. How does volume scale under a transformation in $\mathbb{R}^3$?
- 42. What is the Jacobian determinant for spherical coordinates?
- 43. What is the Jacobian determinant for cylindrical coordinates?

9.8 Deep Understanding / Proof-Level Questions

- 44. How can the change of variables theorem be justified using Riemann sums?
- 45. Why does the determinant appear naturally in linear approximations?
- 46. How does the change-of-variables formula relate to linear algebra?
- 47. In what sense is the Jacobian determinant a multidimensional generalization of dx/du ?
- 48. Why is differentiability essential in the theorem?
- 49. What would fail if the transformation were not invertible?
- 50. How does this theorem connect to the substitution rule in single-variable calculus?

10 Jacobian Matrix and Determinant

10.1 Basic Definitions

1. Let $F : \mathbb{R}^n \rightarrow \mathbb{R}^m$ be given by

$$F(x_1, \dots, x_n) = (f_1(x), \dots, f_m(x)).$$

What is the definition of the *Jacobian matrix* $J_F(x)$?

2. Write explicitly the Jacobian matrix:

$$J_F(x) = \left(\frac{\partial f_i}{\partial x_j} \right).$$

What are its dimensions?

3. In which case is the Jacobian matrix square?
4. When the Jacobian matrix is simply the transpose of the gradient vector?
5. When $m = n$, how is the *Jacobian determinant* defined?
6. What does the notation

$$\det J_F(x)$$

represent?

10.2 Jacobian as Derivative

7. How is the Jacobian matrix related to the total derivative (Fréchet derivative) of a function $F : \mathbb{R}^n \rightarrow \mathbb{R}^m$?
8. State the linear approximation formula:

$$F(x + h) \approx F(x) + J_F(x)h.$$

9. What does the Jacobian matrix represent geometrically at a point?
10. What condition on $J_F(x)$ ensures that F is locally invertible near x ?

10.3 Chain Rule

11. Let $F : \mathbb{R}^n \rightarrow \mathbb{R}^m$ and $G : \mathbb{R}^m \rightarrow \mathbb{R}^p$. State the multivariable chain rule for Jacobians:

$$J_{G \circ F}(x) = ?$$

12. When $n = m = p$, what is the corresponding formula for determinants?
13. Why does this imply

$$\det J_{G \circ F}(x) = \det J_G(F(x)) \cdot \det J_F(x)?$$

10.4 Geometric Interpretation

14. How does the Jacobian determinant describe local volume scaling?
15. What does it mean if $\det J_F(x) > 0$?
16. What does it mean if $\det J_F(x) < 0$?
17. What does it mean if $\det J_F(x) = 0$?
18. How does the absolute value $|\det J_F(x)|$ relate to infinitesimal volume elements?

10.5 Change of Variables in Integration

19. State the change-of-variables formula in $\mathbb{R}^n$:

$$\int_{F(U)} g(y) dy = \int_U g(F(x)) |\det J_F(x)| dx.$$

20. Why is the absolute value necessary?
21. What conditions must F satisfy for the change-of-variables formula to hold?
22. Compute the Jacobian determinant for the transformation from Cartesian to polar coordinates:

$$x = r \cos \theta, \quad y = r \sin \theta.$$

23. Why does the area element become

$$dx dy = r dr d\theta?$$

10.6 Special Cases and Examples

24. Compute the Jacobian matrix of

$$F(x, y) = (x^2 y, e^{xy}).$$

25. Compute the Jacobian determinant of

$$F(x, y) = (u, v)$$

where

$$u = x + y, \quad v = x - y.$$

26. For a linear transformation $F(x) = Ax$, what is $J_F(x)$?
27. How is $\det J_F(x)$ related to $\det A$ in the linear case?
28. If F is a rotation in $\mathbb{R}^2$, what is its Jacobian determinant?

10.7 Inverse Function Theorem

29. State the inverse function theorem in terms of the Jacobian determinant.
30. Why does $\det J_F(x_0) \neq 0$ imply local invertibility?
31. How is the Jacobian of the inverse function expressed in terms of J_F ?

10.8 Higher Concepts

32. How does the Jacobian generalize the derivative of a single-variable function?
33. In what sense is the Jacobian matrix the best linear approximation?
34. How does the determinant relate to orientation?
35. What is the Jacobian in the context of manifolds and coordinate changes?
36. How does the Jacobian relate to differential forms?

11 Hessian Matrix

11.1 Basic Definitions

1. What is the Hessian matrix of a scalar-valued function $f : \mathbb{R}^n \rightarrow \mathbb{R}$?
2. Write the general form of the Hessian matrix for a function $f(x_1, x_2, \dots, x_n)$.
3. What are the entries of the Hessian matrix in terms of partial derivatives?
4. For $f(x, y)$, write the Hessian explicitly.

11.2 Computation

5. Compute the Hessian of $f(x, y) = x^2 + y^2$.
6. Compute the Hessian of $f(x, y) = x^2y + y^3$.
7. Compute the Hessian of $f(x, y, z) = xyz$.
8. Compute the Hessian of $f(x, y) = e^{x^2+y^2}$.

11.3 Interpretation

9. What information does the Hessian matrix encode about a function?
10. How is the Hessian related to the second-order Taylor expansion?
11. What does the Hessian tell you about local curvature?

11.4 Critical Points and Optimization

12. What is a critical point of a multivariable function?
13. State the second derivative test using the Hessian for functions of two variables.
14. What are the conditions on the Hessian for:
 - a local minimum?
 - a local maximum?
 - a saddle point?
15. What does it mean for the Hessian to be positive definite?
16. What does it mean for the Hessian to be negative definite?
17. What happens if the Hessian is indefinite?
18. What happens if the determinant of the Hessian is zero?

11.5 Matrix Properties

19. When is the Hessian symmetric?
20. State the conditions under which mixed partial derivatives are equal.
21. Why is symmetry of the Hessian important?

11.6 Eigenvalues and Geometry

22. How are eigenvalues of the Hessian related to curvature?
23. What does it mean if all eigenvalues are positive?
24. What does it mean if eigenvalues have mixed signs?
25. How do eigenvectors of the Hessian relate to principal directions?

11.7 Advanced Connections

26. How does the Hessian relate to convexity of a function?
27. State the condition for a function to be convex using the Hessian.
28. What is the Hessian in the context of optimization algorithms (e.g., Newton's method)?
29. How is the Hessian used in quadratic approximation?
30. What is the bordered Hessian and when is it used?

11.8 Conceptual Checks

31. Why does the first derivative test fail for multivariable functions without second-order information?
32. Can a function have a zero gradient but not be an extremum? Explain using the Hessian.
33. How does the Hessian generalize the second derivative from single-variable calculus?

12 Frenet–Serret Frame and Formulas

1. What is a regular curve? What condition must a parametrized curve $\mathbf{r}(t)$ satisfy?
2. What is arc-length parametrization? How do we define the arc-length parameter s ?
3. Why is arc-length parametrization especially convenient when defining the Frenet frame?
4. How is the unit tangent vector $\mathbf{T}(s)$ defined in terms of $\mathbf{r}(s)$?
5. Why does $\|\mathbf{T}(s)\| = 1$ always hold?
6. Show that $\mathbf{T}'(s)$ is orthogonal to $\mathbf{T}(s)$. Why must this be true?
7. How is curvature $\kappa(s)$ defined in terms of $\mathbf{T}'(s)$?
8. What is the geometric meaning of curvature? What does it measure intuitively?
9. Under what condition is curvature zero? What does this imply about the curve?
10. How is the principal normal vector $\mathbf{N}(s)$ defined?
11. Why is $\mathbf{N}(s)$ a unit vector?
12. What is the binormal vector $\mathbf{B}(s)$ and how is it defined?
13. Why is $\mathbf{B}(s)$ orthogonal to both $\mathbf{T}(s)$ and $\mathbf{N}(s)$?
14. Show that $\{\mathbf{T}, \mathbf{N}, \mathbf{B}\}$ forms an orthonormal basis.
15. What is the geometric interpretation of the Frenet frame at a point on the curve?
16. Write down the Frenet–Serret formulas.
17. What does the equation $\mathbf{T}'(s) = \kappa(s)\mathbf{N}(s)$ tell us geometrically?
18. How is torsion $\tau(s)$ defined?
19. What is the geometric meaning of torsion?
20. What does it mean if torsion is zero everywhere?
21. Explain the equation $\mathbf{B}'(s) = -\tau(s)\mathbf{N}(s)$.
22. Why must $\mathbf{N}'(s)$ involve both $\mathbf{T}(s)$ and $\mathbf{B}(s)$?
23. Derive the Frenet–Serret formulas starting from orthonormality constraints.
24. How do curvature and torsion uniquely determine a curve (up to rigid motion)?
25. What is the fundamental theorem of space curves?
26. Compute κ and τ for a straight line.
27. Compute κ and τ for a circle of radius R .
28. Compute κ and τ for a helix.
29. How do you compute curvature for a curve given in a non arc-length parameter t ?
30. What is the formula for curvature using $\mathbf{r}'(t)$ and $\mathbf{r}''(t)$?
31. How is torsion computed using $\mathbf{r}'(t)$, $\mathbf{r}''(t)$, and $\mathbf{r}'''(t)$?
32. Why does the cross product $\mathbf{r}'(t) \times \mathbf{r}''(t)$ appear in curvature and torsion formulas?
33. What happens to the Frenet frame at points where $\kappa = 0$?
34. Can the Frenet frame always be defined globally along a curve?

35. What is the osculating plane? Which vectors span it?
36. What is the normal plane? Which vectors span it?
37. What is the rectifying plane? Which vectors span it?
38. How does the Frenet frame describe the local geometry of a curve?
39. In what sense does curvature measure bending and torsion measure twisting?
40. How would you reconstruct a curve from given curvature and torsion functions?
41. What role do initial conditions play in reconstructing a curve?
42. How does the Frenet frame evolve along the curve?
43. What is the matrix form of the Frenet–Serret equations?
44. Why is the Frenet–Serret matrix skew-symmetric (up to signs)?
45. How is the Frenet frame related to rigid body motion along a curve?
46. What is the connection between angular velocity and the Frenet–Serret formulas?
47. How can the Frenet frame be interpreted as a moving coordinate system?
48. What breaks down if the curve is not twice differentiable?

13 Geometry of Differential Equations

Let's develop a geometric understanding of differential equations by connecting them to vector fields and tangent structures. Instead of viewing a differential equation purely as an algebraic object, we interpret it as a rule that assigns a direction (velocity) to every point in space. This perspective reveals that solutions to differential equations are not arbitrary functions, but special curves—called integral curves—that follow these directions at every point.

We begin by relating slope fields and vector fields, showing that a first-order differential equation naturally defines a direction field. From there, we introduce the concept of integral curves as trajectories that are everywhere tangent to the vector field. This provides a unifying viewpoint: solving a differential equation means finding curves whose tangent vectors agree with a prescribed field.

We then connect these ideas to differential geometry by interpreting vector fields as tangent vector fields and surfaces as collections of tangent planes. This highlights the distinction between local information (given by the vector field) and global behavior (the resulting trajectories).

Finally, we interpret the Forward Euler method geometrically as a procedure that approximates integral curves by taking small steps in the direction of the vector field. This reveals numerical methods as discrete versions of continuous geometric motion.

Overall, the goal is to build an intuitive and structural understanding of differential equations as geometry: vector fields define motion, and solutions arise by following that motion.

13.1 Vector Fields and Differential Equations

1. What does a vector field $\mathbf{V}(\mathbf{x})$ assign to each point in space?
2. In what sense does a vector field define a “local rule of motion”?
3. How can a differential equation $\frac{d\mathbf{x}}{dt} = \mathbf{V}(\mathbf{x})$ be interpreted geometrically?
4. Why does a vector field determine velocity but not a global destination?
5. How does an initial condition $\mathbf{x}(0) = \mathbf{x}_0$ select a unique trajectory?
6. What does it mean for the family of all solutions to be “encoded” in the vector field?

13.2 Slope Fields and Direction Fields

7. How does the equation $\frac{dy}{dx} = f(x, y)$ define a slope field?
8. How can a slope field be converted into a vector field using $(1, f(x, y))$?
9. Why is a slope field really a direction field rather than a unique vector field?
10. How are solution curves related to the directions shown in a slope field?
11. Why can we scale vectors (e.g. $(\lambda, \lambda f)$) without changing the solutions?

13.3 Integral Curves

12. What is the definition of an integral curve in terms of $\gamma'(t) = \mathbf{V}(\gamma(t))$?
13. Why is an integral curve the same thing as a solution to a differential equation?
14. What does it mean geometrically that a curve “follows” a vector field?

15. Why are integral curves also called trajectories or flow lines?
16. How can an integral curve be written in integral form:

$$\gamma(t) = \gamma(0) + \int_0^t \mathbf{V}(\gamma(s)) ds?$$

17. Why is the term “integral” used in “integral curve”?

13.4 Tangent Vectors and Tangent Vector Fields

18. What is a tangent vector to a curve?
19. What is the tangent plane $T_p S$ at a point on a surface?
20. Why does a tangent plane contain infinitely many possible tangent directions?
21. What is a tangent vector field on a surface S ?
22. Why does a tangent vector field require choosing one vector from each tangent plane?
23. How do parametrizations $\mathbf{r}(u, v)$ produce tangent vectors $\mathbf{r}_u$ and $\mathbf{r}_v$?
24. How can a general tangent vector be expressed as a linear combination $a\mathbf{r}_u + b\mathbf{r}_v$?
25. How can a vector field in $\mathbb{R}^3$ be projected onto a tangent plane?
26. How do curves on a surface naturally generate tangent vector fields?

13.5 Connection Between Tangent Fields and Differential Equations

27. How does the condition $\dot{\gamma}(t) = \mathbf{V}(\gamma(t))$ relate tangent vectors to vector fields?
28. Why can solution curves be viewed as curves whose tangent vectors match a vector field?
29. In what sense is a differential equation prescribing a tangent vector field?
30. How does this interpretation unify slope fields, vector fields, and tangent vectors?

13.6 Forward Euler Method

31. What is the Forward Euler update rule:

$$\mathbf{x}_{n+1} = \mathbf{x}_n + h\mathbf{V}(\mathbf{x}_n)?$$

32. How can this be interpreted as “taking a small step in the direction of the vector field”?
33. Why does Forward Euler approximate integral curves with piecewise linear segments?
34. What assumption does Euler make about the vector field over a small time step?
35. How does the exact solution use the integral:

$$\mathbf{x}(t+h) = \mathbf{x}(t) + \int_t^{t+h} \mathbf{V}(\mathbf{x}(s)) ds?$$

36. Why is Forward Euler equivalent to a left-endpoint Riemann sum?
37. Why does error accumulate in Forward Euler when the vector field changes along the path?

13.7 Local vs Global Behavior

- 38. What is the difference between a local rule (vector field) and a global object (solution curve)?
- 39. Why does repeatedly applying local information produce a global trajectory?
- 40. How does this idea relate to numerical methods like Forward Euler?

13.8 Flows and Deeper Geometric View

- 41. What is a flow $\varphi_t(\mathbf{x}_0)$ generated by a vector field?
- 42. How does a vector field define motion over time through its integral curves?
- 43. How does Forward Euler approximate a continuous flow with a discrete map?
- 44. Why can differential equations be viewed as describing motion on geometric spaces?
- 45. How does restricting motion to a surface require the vector field to lie in the tangent plane?